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SHORT REPORT OPEN ACCESS

Short Report: An Examination of Children's Autism Traits and Their Association With Family Experience

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ABSTRACT

The present study examined how the frequency and impact of autism traits were associated with parent-reported family experience, with differential consideration of parental experience and overall family life. Prior research has found an association between autism traits and family functioning; however, less is known about how the impact and frequency of these traits influence family experience. We analyzed data from families of 206 young autistic children (144 boys and 62 girls), aged 2–7, using the Autism Impact Measure (AIM) and the Autism Family Experience Questionnaire (AFEQ). Hierarchical regression analysis showed that the impact of autistic traits was significantly associated with parent experience scores and that the frequency of autistic traits was significantly associated with family life scores in the first-year post-diagnosis. Neither child age nor gender was significantly associated with parent experience or family life. This study highlights the importance of tailoring post-diagnosis family supports to the child's specific features and calls attention to areas for future investigation on this topic, including the exploration of longitudinal trends or demographic factors that may affect parent experience and family life.

1 | Introduction

Autism spectrum disorder (hereafter, autism) is a neurodevelopmental condition characterized by social differences and specialized or intense interests and behaviors (American Psychiatric Association 2022). Globally, estimates of autism prevalence have increased in recent years, which has been attributed in part to greater awareness and support efforts for autistic individuals (Zeidan et al. 2022). Many families make adjustments in their day-to-day lives to better support their autistic child.

Family experience encompasses many elements of the family, such as a family's ability to respond to stress, quality of relationships within the family, and ease of everyday life (Sikora et al. 2013). For the purpose of the present study, family experience is divided into parent experience and family life as a whole, based on the subscales as found in the Autism Family Experiences Questionnaire (AFEQ; Leadbitter et al. 2017). The parent experience involves several factors, including parental confidence, coping strategies, capacity for helping their child or children, and belief in one's abilities as a parent (Leadbitter et al. 2017). Family life involves the overall well-being of one's family system, which may be affected by conflict or stress (Leadbitter et al. 2017). Autistic children and youth often present with challenges in functional domains, requiring parents to delegate additional energy to properly support their child. Parents may also pursue intervention services to support their child's development, which can be both time-intensive and financially stressful (Factor et al. 2019).

A recent review by Desquenne Godfrey et al. (2023) found that families with an autistic child or children generally experience more challenges than families whose members are all neurotypical. In this review, family difficulties were correlated with greater caregiver demand and reduced access to resources. The authors also noted that family life is impacted by autism, as families with an autistic child show more evidence of impaired close relationships within the family.

Kishimoto et al. (2023) investigated how autism traits and family dynamics influence parental stress for caregivers of autistic children. Within family dynamics, the authors explored family adaptability (capacity to adapt to stressors) and cohesion (family support and bonds). Results indicated that parental stress and the impact of autistic traits are positively correlated, suggesting that families with children who require substantial support due to functional challenges experience greater family difficulties, and thus, greater parental stress. This study also found that family adaptability and cohesion were negatively correlated with trait impact—but positively correlated with parental self-efficacy. The authors proposed that parental self-efficacy is protective against parental stress, regardless of trait impact.

Although the relationship between autism traits and family experience has been explored in past studies, it is unclear how family life and parent experience may be affected by autistic trait frequency and impact. While past studies have broadly explored how having an autistic child compares to raising neurotypical children, the specific role of autism traits on family life and parent experience remains unclear (Desquenne Godfrey et al. 2023; Herring et al. 2006; Koziarz et al. 2021). Understanding this association is of particular interest for improving both child and family outcomes. A reciprocal relationship exists between family experience and child development, as more struggles in family experience can lead to caregiver distress, resulting in weakened relationships between caregivers and their children, and thus lower child well-being (Lindsey and Barry 2018). Furthermore, lower parental confidence has been linked to more problematic behaviors in children due to ineffectual parenting practices, such as inconsistent discipline (Lindsey and Barry 2018). Understanding these relationships is important to reveal targets for support for autistic children, their parents, and other family members.

The current study aimed to explore the influence of frequency and impact of autism traits on family experience, teasing apart parent experience and family life. Based on past research, we hypothesized that higher levels of autism trait frequency and impact would be associated with more difficulties in parent experience and family life (Desquenne Godfrey et al. 2023; Kishimoto et al. 2023).

2 | Method

2.1 | Participants

There were 206 families of autistic children aged 2–7 years included in the present study. Families were drawn from the Pediatric Autism Research Cohort (PARC) study which recruits families with autistic children who received their formal diagnosis within the 12 months preceding their enrollment in the longitudinal study (Kata et al. 2024). The PARC study recruits participants from a variety of places including diagnostic centers, service providers, community events, parent support groups, and word of mouth. The only exclusion criteria are not being between the ages of 2–7, parents not being fluent in English (or Hebrew in Israel), and receiving an autism diagnosis longer than one year ago. Participants were recruited at six sites across Canada and one site in Israel. All participants provided informed consent, and the study was approved by the ethics boards at all sites. Demographic information can be found in Table 1.

2.2 | Measures

2.2.1 | Autism Traits

Autism traits were measured using the 41-item Autism Impact Measure (AIM, Mazurek et al. 2020). The AIM is comprised of five domains: Repetitive Behavior, Atypical Behaviors, Communication, Social Reciprocity, and Peer Interactions (Mazurek et al. 2020). On each item, parents rank the trait frequency from 1 (*Never*) to 5 (*Always*) based on how often they have observed the trait in the last 2 weeks, and then the impact of the trait is ranked from 1 (*Not at all*) to 5 (*Severely*) based on how much the trait impacts daily experience. Frequency and Impact are rated independently by the parent and both of these scales are used in the current analysis. Internal consistency was strong for both Frequency measures ($\omega = 0.91$) and Impact measures ($\omega = 0.96$).

2.2.2 | Family Experience

Family experience was measured using the AFEQ (Leadbitter et al. 2017). This 48-item measure is divided into 4 domains. In the present study, we only used items from 2 domains: The Experience of Being a Parent of a Child with Autism (13 items) and Family Life (9 items). The other domains measure child-specific factors which were not the focus of the current study. On each item, parents rank their responses from 1 (*Always*) to 5 (*Never*). A sample item from the Experience of Being a Parent of a Child with Autism domain is *I have coping mechanisms to help my child*. A sample item from the Family Life domain is *Family life is calm*. Internal consistency was strong for Parent Experience ($\omega = 0.90$) and Family Life ($\omega = 0.89$).

TABLE 1 | Demographic characteristics of the participants.

Sample characteristics	*n*	%	*M*	SD
Age			3.89	1.01
Gender of child				
Male	144	70		
Female	62	30		
Responder relationship to child				
Mother	180	87.38		
Father	19	9.22		
Grandparent	2	0.97		
Aunt	1	0.48		
Unspecified parent	2	0.97		
Missing	2	0.97		
Responder ethnicity				
White	130	63.11		
East Asian	6	2.91		
Southeast Asian	8	3.88		
South Asian	8	3.88		
West Asian	2	0.97		
Arab	9	4.37		
Black	15	7.28		
Latin American	7	3.40		
Aboriginal/native	11	5.34		
Other	7	3.40		
Missing data	3	1.46		
Total family annual income				
Less than 10,000	9	4.37		
10,000–14,999	9	4.37		
15,000–19,999	9	4.37		
20,000–29,999	9	4.37		
30,000–39,999	11	5.34		
40,000–49,999	10	4.85		
50,000–59,999	7	3.40		
60,000–69,999	13	6.31		
70,000–79,999	14	6.80		
80,000–89,999	8	3.88		
90,000–99,999	17	8.25		
100,000–124,999	24	11.65		
125,000–149,999	16	7.77		
150,000–199,999	25	12.14		
200,000 or more	14	6.80		
Don't know	7	3.40		
Refusal	3	1.46		
Missing data	1	0.48		

(Continues)

TABLE 1 | (Continued)

Note: N = 206.

2.3 | Procedure

Parents participating in the PARC study completed online questionnaires regarding their child, family, and personal experiences with support services for autism (Kata et al. 2024). The parents completed the questionnaires once every 6 months until their child reached 8 years of age (Kata et al. 2024). This study includes baseline data only.

2.4 | Data Analysis

Two hierarchical regression analyses were used to investigate how parent-reported overall trait frequency and overall trait impact (separately) were associated with scores on the two subscales of the AFEQ, with age and gender as covariates. All continuous predictor variables were mean-centered to ensure interpretability. All assumptions necessary for a hierarchical regression analysis were tested and met.

3 | Results

Descriptive statistics are presented in Table 2. Bivariate correlations can be found in Table 3.

A first hierarchical linear regression was performed to investigate the association between trait frequency and trait impact, and parent experience. This was tested using two models. Results of the analyses can be found in Tables 4 and 5.

Model 1 included age and gender as predictors, with parent experience as the outcome variable. Age and gender showed no significant association with family experience scores, all p's > 0.63.

Model 2 included trait frequency and trait impact as predictors, along with age and gender as covariates, and used parent experience as the outcome variable. Trait impact ($p = 0.002$) showed significant associations with the parent experience score. As autistic trait impact increased, parent experience difficulties increased. The results for Model 2 indicate that the inclusion of trait frequency and trait impact scores accounts for 14% of the variance in parent experience scores.

Model 3 included age and gender as predictors, with family experience as the outcome variable. Age and gender showed no significant association with family experience scores, all p's > 0.20.

Model 4 included trait frequency and trait impact as predictors, along with age and gender as covariates, and used family life as the outcome variable. Trait frequency ($p < 0.001$) showed significant associations with the family life score. As autistic trait frequency increased, family experience difficulties increased. The results for Model 4 indicate that the inclusion of trait frequency and trait impact scores accounts for 22% of the variance in family life scores.

TABLE 2 | Descriptive statistics.

	Minimum	Maximum	Mean	SD
Child age	2	7	3.89	1.01
AIM frequency score	78	182	126.25	21.39
AIM impact score	41	187	97.86	28.59
Family life score	28	93	60.13	12.40
Parent experience score	14	57	35.06	7.71

Note: $N = 206$.

TABLE 3 | Bivariate correlations.

Measure	1	2	3	4
1. Age	—			
2. Frequency score	−0.05	—		
3. Impact score	−0.04	0.68***	—	
4. Family experience score	0.06	0.41***	0.40***	—

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

4 | Discussion

The present study examined the association of child autistic traits (frequency and impact) with parental experience and family life. We found that age and gender were not significantly associated with parental and family difficulties. However, the addition of AIM scores for frequency and impact of traits considerably improved the model's ability to explain variance in AFEQ scores relating to parent and family difficulties, respectively. Overall, greater trait impact was associated with greater parent experience difficulties, and greater trait frequency was associated with greater family life difficulties. It is of considerable notice that despite the intercorrelation between trait impact and frequency, the analyses were able to specifically distinguish two separate contributions for parent and whole family experience, respectively.

The above findings are consistent with previous research stating that families with an autistic child generally experience more difficulties in family experience (Sikora et al. 2013; Desquenne Godfrey et al. 2023). Durán-Pacheco et al. (2022) found that autism trait impact and frequency were related to greater caregiver strain or parental burden, which may include poorer mental and physical health outcomes or relational or financial troubles. Our findings extend this work by suggesting that the impact of autistic traits specifically contributes to the experience of being a parent of an autistic child and that trait frequency may put a strain on overall family life.

These findings may be useful in identifying families with autistic children who require extra support in the months following the diagnosis. It is possible that parental stress levels,

TABLE 4 | Hierarchical regressions for parent experience scores.

Variable	*B*	SE	*β*	*p*	ΔR^2	Adjusted R^2
Model 1					0.002	−0.008
Constant	35.81	1.62		< 0.001***		
Age	0.19	0.54	0.03	0.73		
Gender	−0.58	1.18	−0.03	0.63		
Model 2					0.136	0.121
Constant	36.57	1.52		< 0.001***		
Age	0.30	0.50	−0.069	0.55		
Gender	−1.16	1.10	0.04	0.29		
Trait frequency	0.04	0.03	0.12	0.18		
Trait impact	0.08	0.02	0.28	0.002**		

Note: Age, frequency, and impact centered.
*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

TABLE 5 | Hierarchical regressions for family experience scores.

Variable	B	SE	β	p	ΔR^2	Adjusted R^2
Model 3					0.015	0.005
Constant	26.43	1.29	0.09	< 0.001***		
Age	0.55	0.43	−0.08	0.20		
Gender	−1.05	0.94		0.27		
Model 4					0.221	0.221
Constant	27.05	1.15		< 0.001***		
Age	0.68	0.38	0.11	0.07		
Gender	−1.52	0.83	−0.11	0.07		
Trait frequency	0.11	0.02	0.40	< 0.001***		
Trait impact	0.02	0.02	0.10	0.22		

Note: Age, frequency, and impact centered.
*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

confidence in parenting, or coping skills in parents of autistic children may reflect the level of support needed for the child. More work is needed to disentangle this relationship and also which specific types of traits are most impactful on the family. Traits are reflected in family experience and may help to inform interventions. Additionally, future investigation into this relationship could help professionals to design supports promoting skill development while enhancing overall family well-being.

Our findings are consistent with a study by McStay et al. (2013) that found child age is not significantly associated with parenting stress for parents of autistic children. Our finding that gender does not have a significant association with family experience conflicts with previous research demonstrating greater parenting stress for parents of autistic daughters as compared to parents of autistic sons (Zamora et al. 2014), which may be a function of the relatively young age of our sample. Our results suggest that, within the first-year post-diagnosis, autism trait impact and the frequency of associated behaviors have more of an impact on the parent and family experience, respectively, than age and gender. The parent and family experience may also be affected by other factors not considered in this study, such as access to services, social support, and experiences of stigma.

This study has some limitations. The PARC study aims to capture a pediatric population, and as such, data collection for participants is discontinued when they reach 8 years of age. While the current study is informative for parents and families of younger autistic children, it is important that further investigation be conducted for families of older autistic children and adolescents. Additionally, the current study only used data from one time-point and does not capture the development of the children or the evolution of family experience. Once completed, our longitudinal study will enable us to explore family experience over time to extend this work and further provide insight into where intervention efforts are most needed to support autistic children and their families. This study also did not consider socioeconomic status and does not reveal how demographic factors outside of age and gender may affect family experience.

A potential important limitation of the present study is the absence of intelligence quotient (IQ) as a covariate in our analysis. Including a measure for intellectual ability as a covariate could have influenced our results, as Desquenne Godfrey et al. (2023) demonstrated in their systematic review that intelligence level is positively associated with family functioning in families of autistic children. Future research may benefit from incorporating IQ as a covariate to better understand the relationship between autism traits and family functioning outcomes.

Another limitation is that most questionnaires were completed by mothers, and findings may therefore primarily reflect maternal perspectives. This pattern is consistent with systematic reviews demonstrating that autism research samples are typically dominated by mothers, with fathers underrepresented (Rankin et al. 2019; Clark-Whitney et al. 2026).

Lastly, because the same parent respondent completed both the AIM and the AFEQ, these findings are vulnerable to shared method variance. This makes it difficult to disentangle whether higher reported autistic traits shape family and parent experiences, or whether parents' perceptions of family stress, burden, or cohesion influence how they interpret and report their child's behaviors. This limitation highlights a need for future research incorporating multi-informant designs to strengthen causal interpretations. In conclusion, findings from the present study emphasize the importance of autistic traits—both in terms of frequency and impact—in predicting parent and family experience in the first-year post-diagnosis. These results are consistent with previous work on how autistic behaviors influence family dynamics. Future directions could include a longitudinal design to reveal how family experience evolves with child development and could highlight targets for intervention as autism traits and behaviors emerge or dissipate with age. Further research is also required to examine the mechanisms of this association—Is it autistic traits themselves or a third variable such as access to services which is driving this association?

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Ethics Statement

This study was approved by all relevant research ethics boards for each site.

Consent

Parents all were engaged in informed consent.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data will be made available upon reasonable request.

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